

# The right page is usually **found**, but rarely **first**.

How the system answers a question today:

- 1 Trained Hebrew model turns the question into a meaning vector
- 2 Elasticsearch returns the **50 closest paragraphs**
- 3 Collapse to pages. A page ranks by its **single best paragraph**

Step 3 is the weak spot. A page with *one* paragraph that happens to look like the question can outrank the page that is actually *about* the question.

FULL CORPUS · 24,487 PARAGRAPHS · BASELINE

Correct page in the **top 10**

**87%**

Correct page at **#1**

**39%**

The right page is already among the candidates for almost 9 questions in 10, but only the top few pages reach the answer step. **That gap is the target.**

# Score the **page**, not just its luckiest paragraph.

```
page score = best paragraph
+ 0.25 × second-best paragraph
+ 0.25 × title match
```

All three from the **same trained model**. A small margin gate (0.05) keeps generic "hub" pages from jumping up from far below.

## Second-best paragraph

The right page usually has several matching paragraphs. A look-alike has one.

## Title match

Kol-Zchut titles name the topic. Paragraphs are fragments that often don't.

עד מתי מקבלים קצבת ילדים? → קצבת ילדים

no new model

no re-indexing

one flag, off = original

milliseconds per query

## WHY NOT THE OBVIOUS UPGRADES?

The shipped retriever was **trained on these very questions**. Anything that overrules it loses. Both ideas were built, integrated and tested the same way.

| IDEA                                                                | HELD-OUT #1, % |
|---------------------------------------------------------------------|----------------|
| <b>Cross-encoder reranker</b> (BGE)<br>re-reads and re-sorts the 50 | 45 → 40        |
| <b>Page scoring</b><br>listens to the same model more carefully     | 39 → 55        |

Held-out sets differ: 100 questions for the reranker, 250 fresh ones for page scoring (on those same 100 it was 45 → 50). The rejected idea helped ranks 4–5 and hurt the first result.

# First result: 38.8% → 54.8% on questions the weights never saw.

| CORRECT PAGE IS...               | BEFORE | AFTER | CHANGE |
|----------------------------------|--------|-------|--------|
| at #1                            | 38.8%  | 54.8% | +16.0  |
| in top 3                         | 65.6%  | 71.6% | +6.0   |
| in top 5                         | 77.2%  | 78.4% | +1.2   |
| in top 10                        | 86.8%  | 87.6% | +0.8   |
| MRR@10 (how high up, on average) | 0.547  | 0.650 | +0.10  |

**Metrics:** hit@k because #1 is what the user and the answer step actually see. MRR because it rewards moving the right page *upward*. Same 50 candidates before and after, only the order changes, so top-5 and top-10 are essentially unchanged (within noise): reordering fixes rank, not recall.

## HOW IT WAS KEPT HONEST

Weights chosen on **250** questions, reported **once** on 250 held-out ones. Ship rule for every idea: held-out #1 or MRR up, top-5 not down. One of two failed it.

## IN THE BACKEND

An optional step in the engine's search, turned on in the local demo. Off restores the original exactly. No API key needed. 28 unit tests, plus a check that the live API returns the offline ranking.

## HONEST LIMITS

The QA file also trained the retriever, so absolute numbers are optimistic. The split guards against tuning, not familiarity. The title weight relies on good titles.